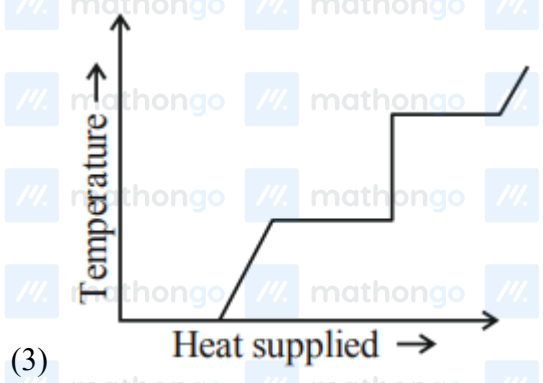
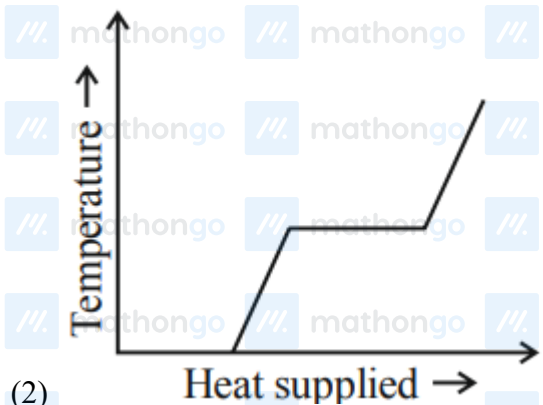
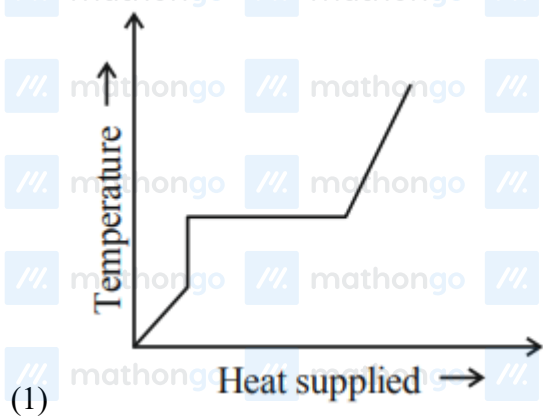


Questions with Answer Keys

MathonGo

Q1 - 2024 (30 Jan Shift 2)

A block of ice at -10°C is slowly heated and converted to steam at 100°C . Which of the following curves represent the phenomenon qualitatively:



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Questions with Answer Keys

MathonGo



Q2 - 2024 (31 Jan Shift 1)

Two conductors have the same resistances at 0°C but their temperature coefficients of resistance are α_1 and α_2 . The respective temperature coefficients for their series and parallel combinations are :

(1) $\alpha_1 + \alpha_2, \frac{\alpha_1 + \alpha_2}{2}$

(2) $\frac{\alpha_1 + \alpha_2}{2}, \frac{\alpha_1 + \alpha_2}{2}$

(3) $\alpha_1 + \alpha_2, \frac{\alpha_1 \alpha_2}{\alpha_1 + \alpha_2}$

(4) $\frac{\alpha_1 + \alpha_2}{2}, \alpha_1 + \alpha_2$

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Questions with Answer Keys

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Answer Key

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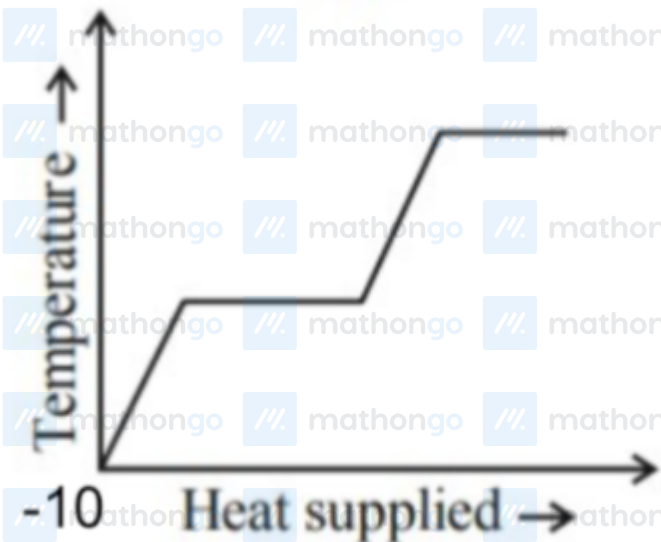
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Solutions

MathonGo

Q1

Most of the students will think no option is correct but we will give the benefit of doubt to NTA since no coordinates are mentioned. In transition we should have two horizontal lines and they assumed the starting point is -10 .



Q2

Series :

$$R_{eq} = R_1 + R_2$$

$$2R(1 + \alpha_{eq}\Delta\theta) = R(1 + \alpha_1\Delta\theta) + R(1 + \alpha_2\Delta\theta)$$

$$2R(1 + \alpha_{eq}\Delta\theta) = 2R + (\alpha_1 + \alpha_2)R\Delta\theta$$

Parallel :

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{\frac{R}{2}(1 + \alpha_{eq}\Delta\theta)} = \frac{1}{R(1 + \alpha_1\Delta\theta)} + \frac{1}{R(1 + \alpha_2\Delta\theta)}$$

$$2[(1 + \alpha_1\Delta\theta)(1 + \alpha_2\Delta\theta)]$$

$$= [2 + (\alpha_1 + \alpha_2)\Delta\theta][1 + \alpha_{eq}\Delta\theta]$$

$$2[1 + \alpha_1\Delta\theta + \alpha_2\Delta\theta + \alpha_1\alpha_2\Delta\theta^2]$$

$$= 2 + 2\alpha_{eq}\Delta\theta + (\alpha_1 + \alpha_2)\Delta\theta + \alpha_{eq}(\alpha_1 + \alpha_2)\Delta\theta^2$$

Neglecting small terms

$$2 + 2(\alpha_1 + \alpha_2)\Delta\theta = 2 + 2\alpha_{eq}\Delta\theta + (\alpha_1 + \alpha_2)\Delta\theta$$

$$(\alpha_1 + \alpha_2)\Delta\theta = 2\alpha_{eq}\Delta\theta$$

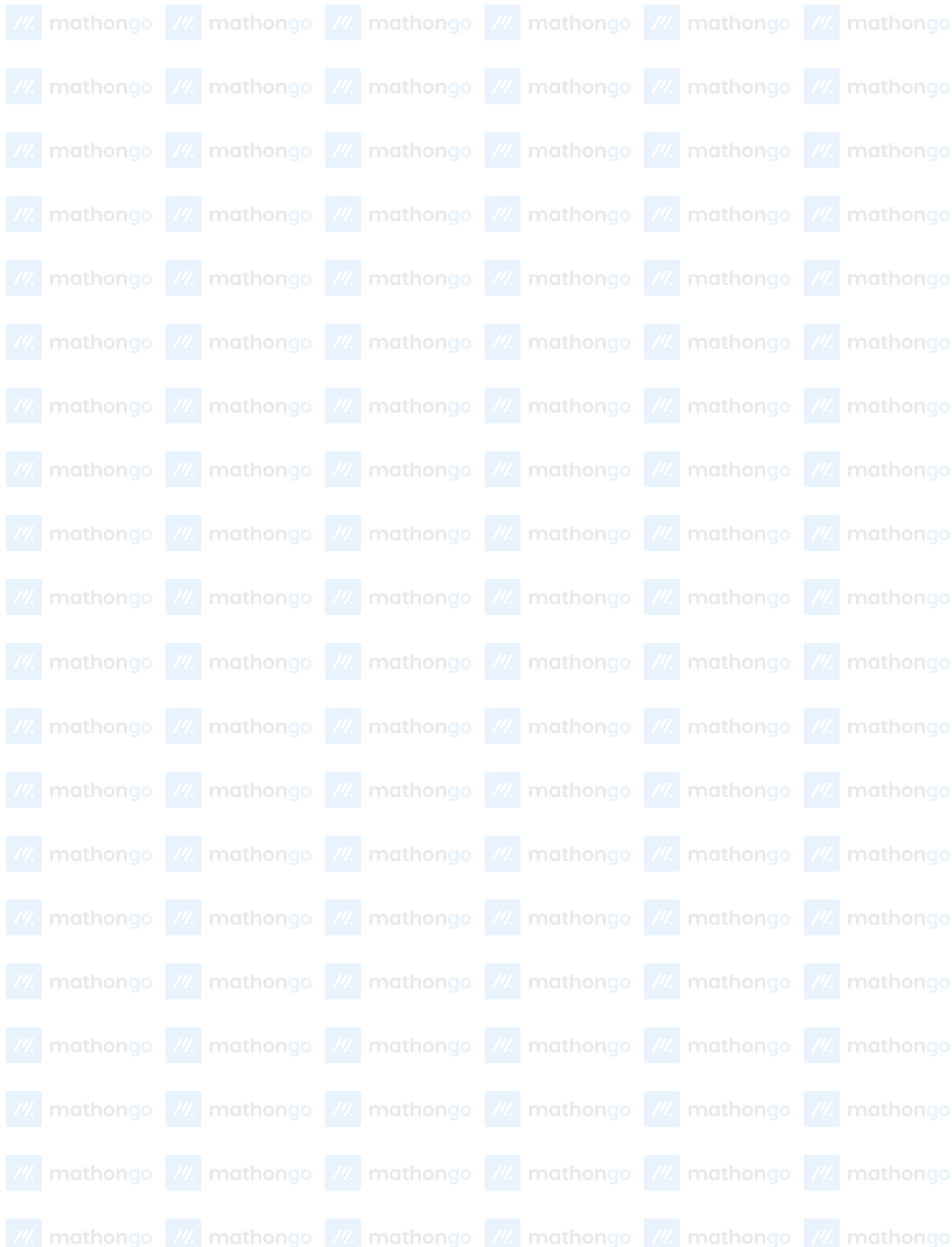
$$\alpha_{eq} = \frac{\alpha_1 + \alpha_2}{2}$$

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Solutions

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